Group Members: Abdalla Elokely, Nolan Berg, Nolan Nachbar, Thomas Hinckley, and Collin Davis

1. Project Domain Selection

Our team has selected Inventory Management as our project domain. Specifically, we are designing a database system for the University of Wyoming IT (UWIT) Classroom Technology Services — the department responsible for deploying, tracking, and maintaining AV and technology equipment in classrooms and learning spaces across campus.

The domain was chosen because all five group members are familiar with the existing system used by UWIT, the Learning Space Manager (LSM), giving us direct access to real requirements, real pain points, and a real organizational context to work from.

2. Organization Overview

2.1 Organization Details

Organization: University of Wyoming IT — Classroom Technology Services

Location: University of Wyoming, Laramie, WY 82071

Current System: Learning Space Manager (LSM) — hosted at uwyo.talem3.com

System Scale: 18,682 individual equipment items tracked across 1,022 campus locations

2.2 What the Organization Does

UWIT Classroom Technology Services is responsible for the installation, maintenance, and lifecycle management of AV and technology equipment in University of Wyoming classrooms, labs, conference rooms, and auditoriums. This includes projectors, displays, audio systems, touchpanels, network switches, computers, microphones, document cameras, and more.

Staff use the LSM system daily to look up equipment by room, track serial numbers, log purchase dates, monitor end-of-life timelines, and identify what catalog items are deployed where across campus.

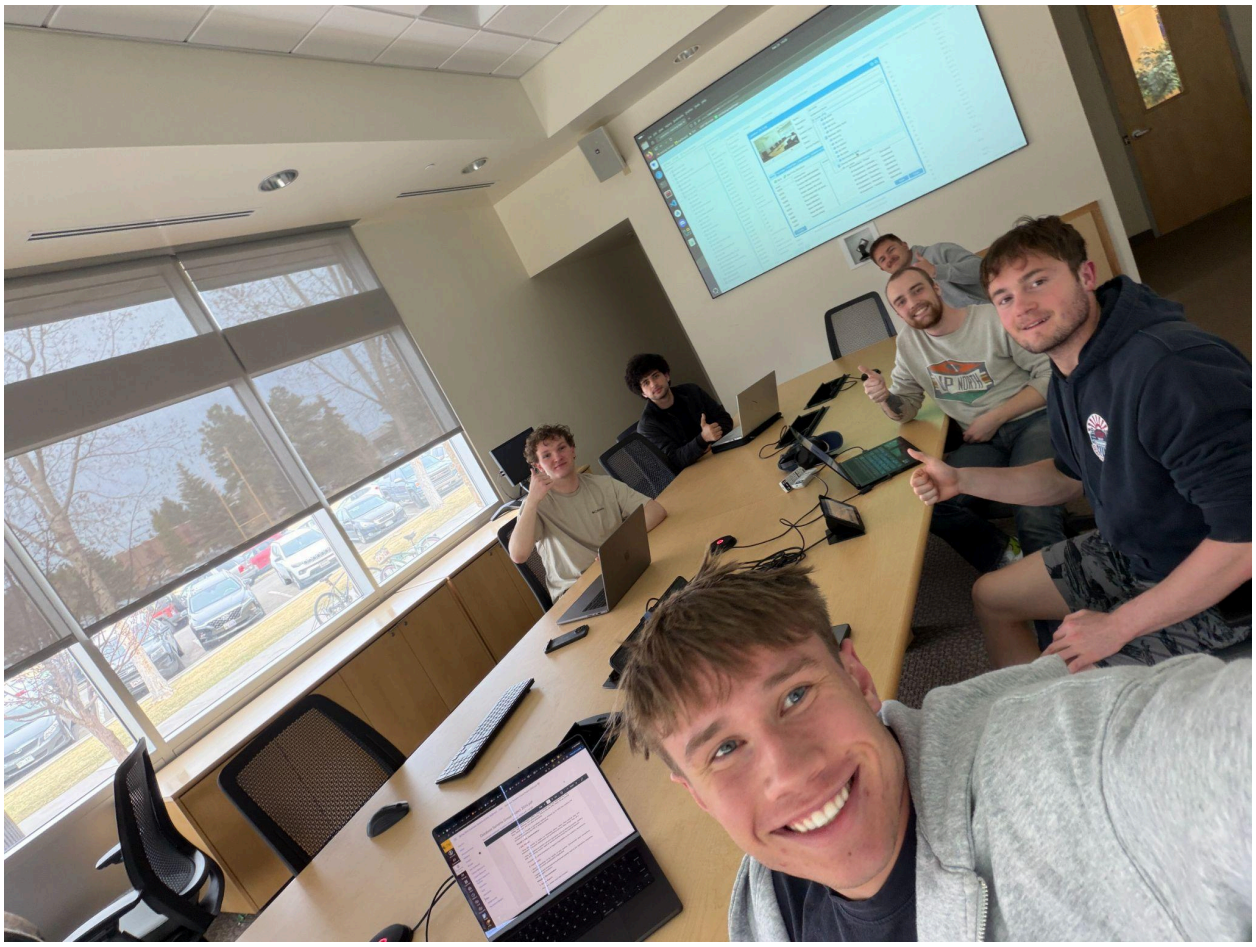
3. Organizational Visit

3.1 How We Conducted Our Visit

Our team conducted our organizational visit with UWIT Classroom Technology Services. Multiple group members are active student employees within the department, working as Classroom Technology Specialists and Inventory Specialists, giving us direct, firsthand access

to the LSM system and the staff who use it daily. We were able to observe the system in active use, consult with department staff about current workflows, and identify pain points from our own experience managing inventory within the system.

Image of meeting with Alex:



3.2 Interview Summary

Through conversations with UWIT staff and our own experience using LSM, we gathered the following key insights:

- The system tracks equipment across three main concepts: where it is (location), what type it is (catalog), and the individual unit itself (item).
- Rooms on campus are organized by building, e.g., WU 0223 means room 0223 in the Wilbur Knight building. The current system does not formally separate building from room, making building-level queries harder.
- Every item has a catalog entry that describes the model and manufacturer. Multiple items of the same model all point to the same catalog record.

- Catalog entries are organized into hierarchical sections, such as All Equipment > Audio > Speakers. This hierarchy is currently stored as a formatted string rather than a relational structure.
- Serial numbers, purchase dates, and MAC addresses (for network equipment) are the most frequently looked-up fields on individual items.
- End-of-life dates are important for budgeting and replacement planning — staff frequently filter for items approaching or past their EOL.
- Staff noted that the current system contains many fields that are never filled in or used, adding visual clutter without adding value.

4. Requirements Gathered

4.1 Data the System Needs to Store

Based on our visit and research, the following data must be captured:

- Buildings: each campus building has a short code (e.g., AG, WU, EN) and a full name.
- Rooms/Locations: each room belongs to a building and has a type (Classroom, Lab, Conference Room, Auditorium), a capacity, and an active/inactive status.
- Catalog Sections: equipment categories organized in a parent-child hierarchy (e.g., All Equipment > Projectors).
- Catalog Entries: each unique equipment model (manufacturer + model number) belongs to a catalog section and has a life expectancy, MSRP, and description.
- Items: individual physical units, each tied to a catalog entry and a location. Key fields include serial number, purchase date, received date, end-of-life date, MAC address (for network devices), and operational status.

4.2 Organizational Goals

UWIT staff identified the following goals for an improved inventory system:

- Quickly look up all equipment in a specific room or building.
- Filter items by status (Operational, In Repair, Retired) to understand current deployment health.
- Identify items approaching or past their end-of-life date for replacement planning.
- Understand how long an item has been deployed (purchase date and received date).
- Browse the equipment catalog by category to see what models are in use across campus.

4.3 Pain Points in the Current System

The following problems were identified in the existing LSM system that our design will address:

- Building and room are combined into a single location field with no relational separation, making building-level queries require string parsing.
- Catalog sections are stored as formatted strings (e.g., 'All Equipment :: Audio :: Speakers') rather than a proper parent-child relational structure.
- Many fields on items are unused or redundant — including duplicate MAC address columns, 'Host Name', 'Reserved for', and 'UW Tag'.
- Catalog entries contain vendor logistics fields (First Ship Date, Last Ship Date, Qty Labor, Unit Labor) that are irrelevant to IT inventory management.
- No distinction between purchase date and received/deployment date, making it difficult to track how long equipment has actually been in service.

5. Entities, Relationships, and Constraints

5.1 Key Entities Identified

- Building: represents a campus building. Attributes: code, full name, and address.
- Location: represents a room within a building. Attributes: room number, type, capacity, status, and owner category.
- Catalog Section: represents an equipment category in a hierarchy. Attributes: name, parent section.
- Catalog: represents an equipment model. Attributes: manufacturer, model, section, status, life expectancy, MSRP, description.
- Item: represents a single physical unit. Attributes: code, serial number, status, purchase date, received date, EOL date, MAC address.

5.2 Key Relationships

- A Building contains many Locations (one-to-many).
- A Location holds many Items (one-to-many).
- A Catalog Section can have many child Catalog Sections (self-referencing one-to-many).
- A Catalog Section contains many Catalog entries (one-to-many).
- A Catalog entry describes many Items (one-to-many).

5.3 Key Constraints

- Every Item must reference a valid Catalog entry: an item cannot exist without knowing what type it is.
- Every Item must reference a valid Location: equipment must be assigned to a room.
- Every Location must reference a valid Building.
- Catalog Section parent_id is nullable: top-level sections have no parent.
- Item code must be unique across the system.

- Item and Catalog status fields are constrained to defined values (e.g., Operational, Retired, In Repair).

6. Project Scope

Our database will focus exclusively on the inventory management side of the LSM system. The following are intentionally out of scope for this project:

- Ticketing and maintenance requests
- Room assessments and LSRS scoring
- User accounts and role-based access
- Vendor and purchase order management

Limiting scope to inventory, locations, catalog, and items allows the team to build a clean, well-normalized system that directly addresses the core data management needs of UWIT without unnecessary complexity.

7. Team & Responsibilities

Each team member will own one table in the database, taking responsibility for its design, SQL implementation, sample data, and queries:

Group Member	Table	Domain
Abdalla Elokely	building	Campus buildings — codes, names, addresses
Nolan Berg	location	Rooms within buildings — type, capacity, status
Nolan Nachbar	catalog_section	Equipment category hierarchy (self-referencing)
Thomas Hinckley	catalog	Equipment models — manufacturer, specs, life expectancy
Collin Davis	item	Individual physical assets — serial numbers, dates, status